



Swiss Institute of  
Bioinformatics

SINGLE-CELL TRANSCRIPTOMICS WITH R

# Integration analysis

Deepak Tanwar  
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# Learning objectives

Understand the importance of experimental design

Identify scenarios where integration is necessary for data analysis

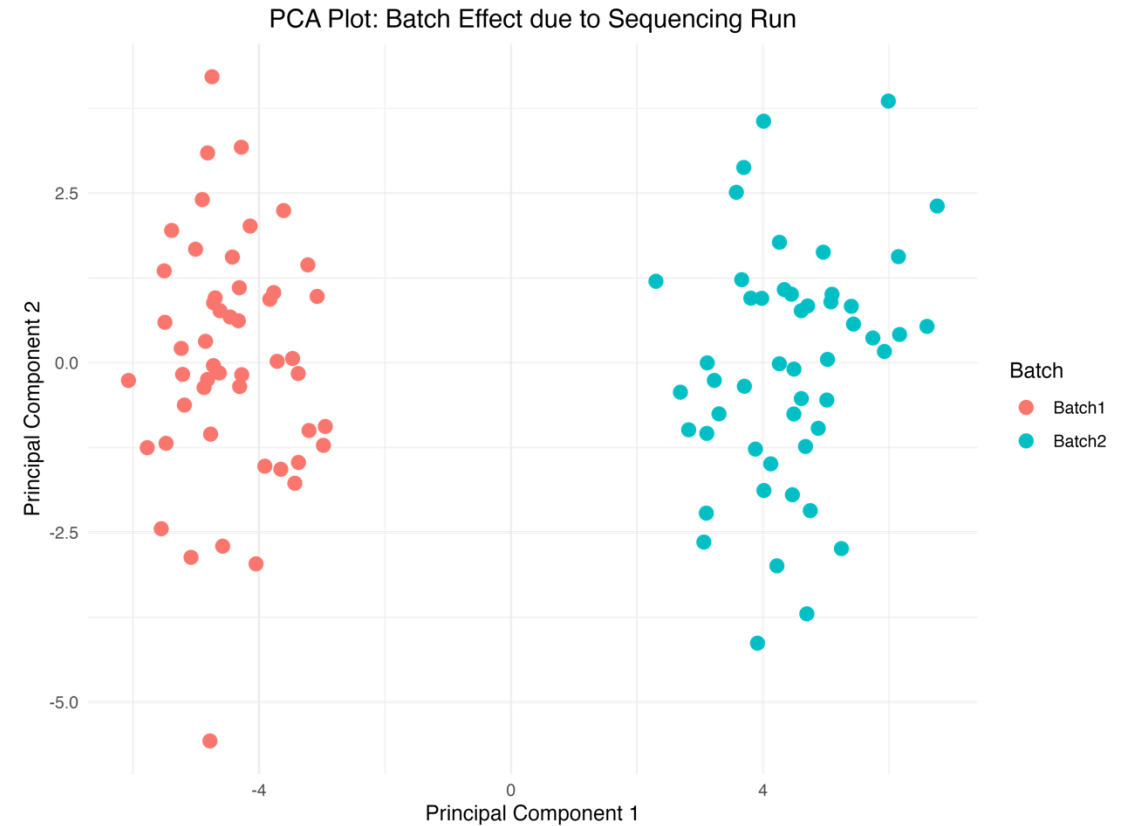
Apply canonical correlation analysis (CCA) for integrating datasets

# Unwanted Sources of Variation

Batch Effect is systematic technical variations due to differences in:

- a) cell isolation and handling protocols,
- b) library preparation technology, and sequencing platforms

Batch effects can obscure true biological signals, making it difficult to compare datasets



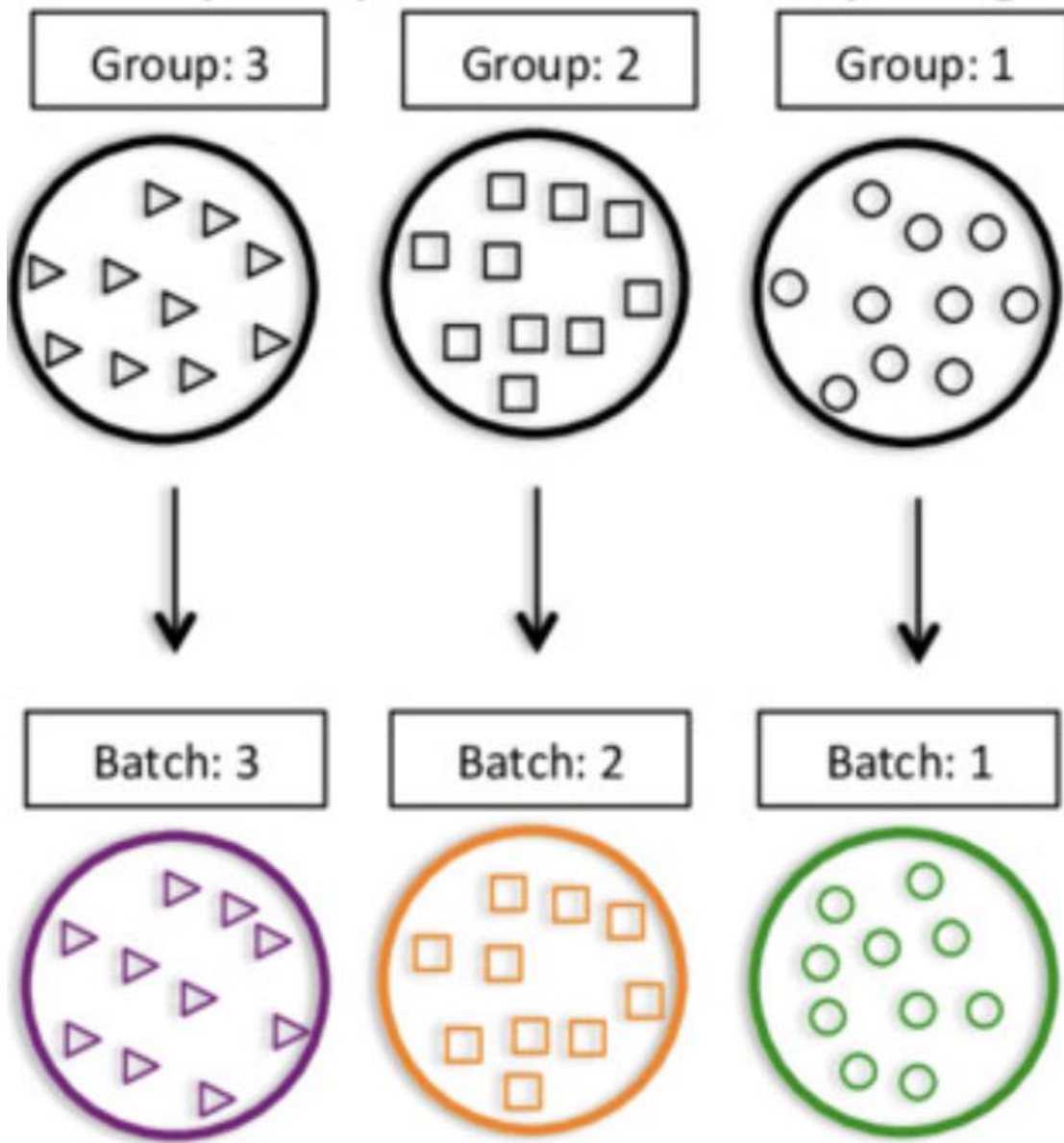
# Unwanted Sources of Variation

Confounders are variables (e.g. Gender, Age) that could influence gene expression

If they are not properly accounted for in the analysis they could potentially lead to misleading associations.

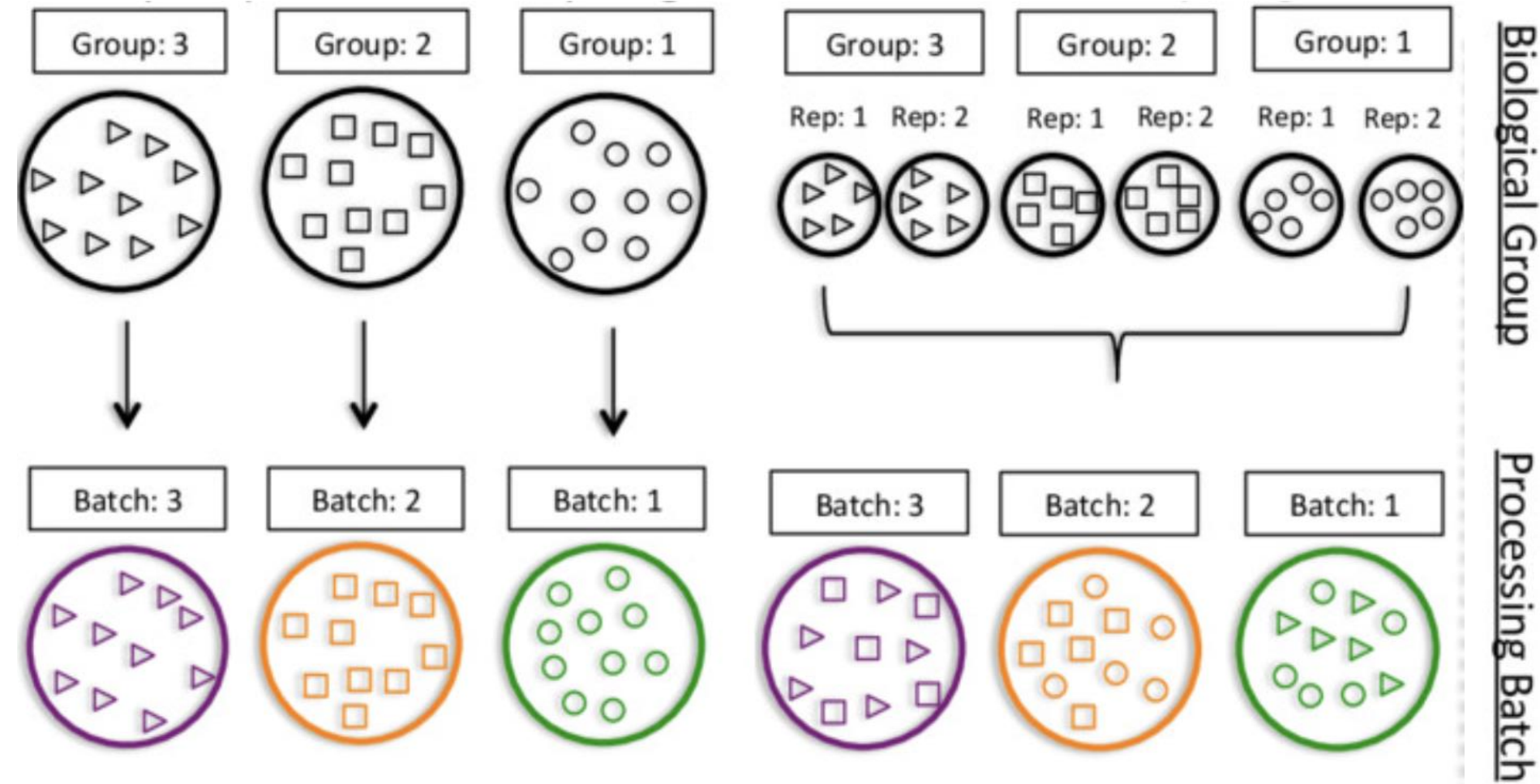


# Experimental design matters

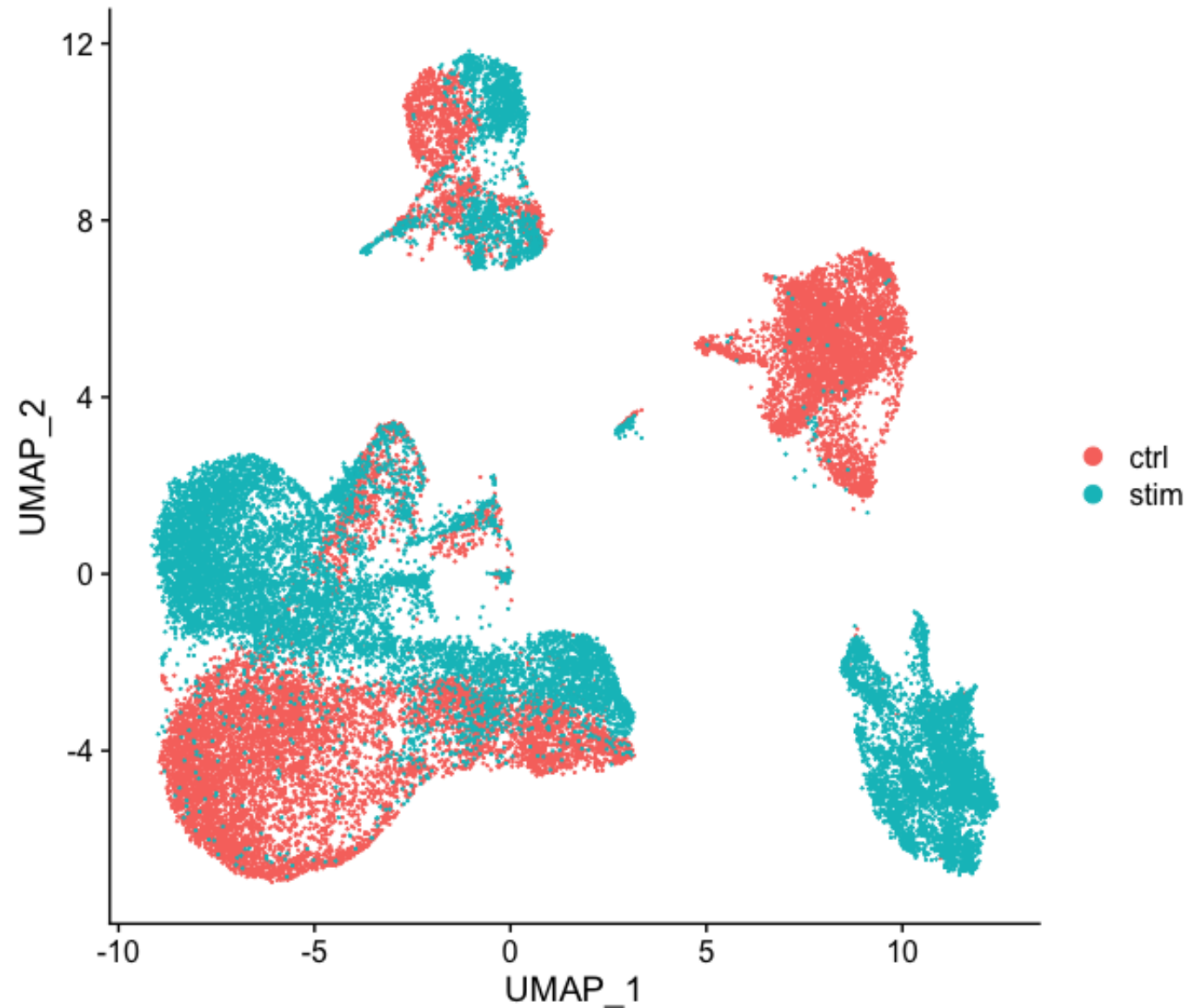


**What changes would you make here to make the experimental design more optimal?**

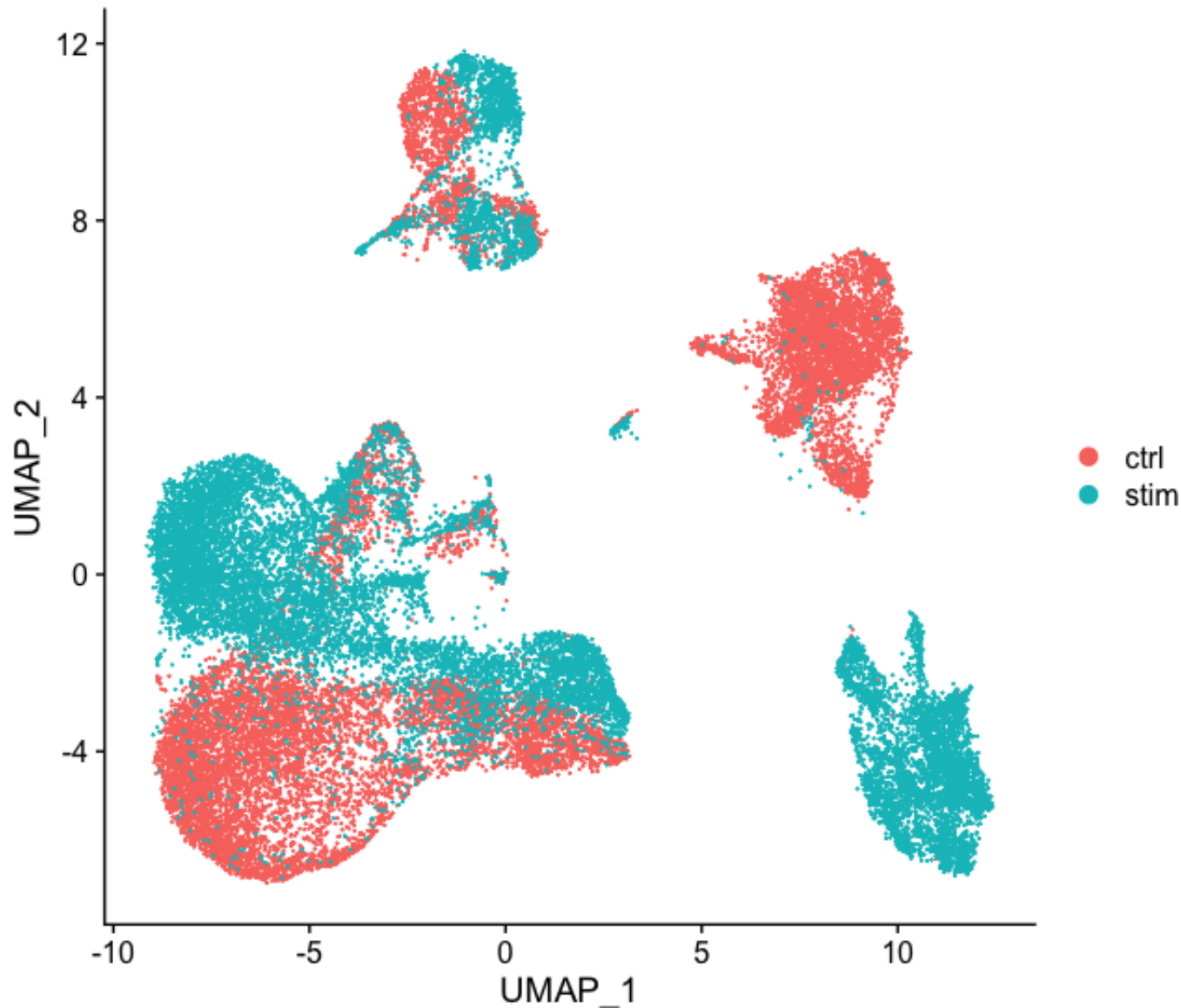
# Experimental design matters



# Exercise: Identify problem in this plot

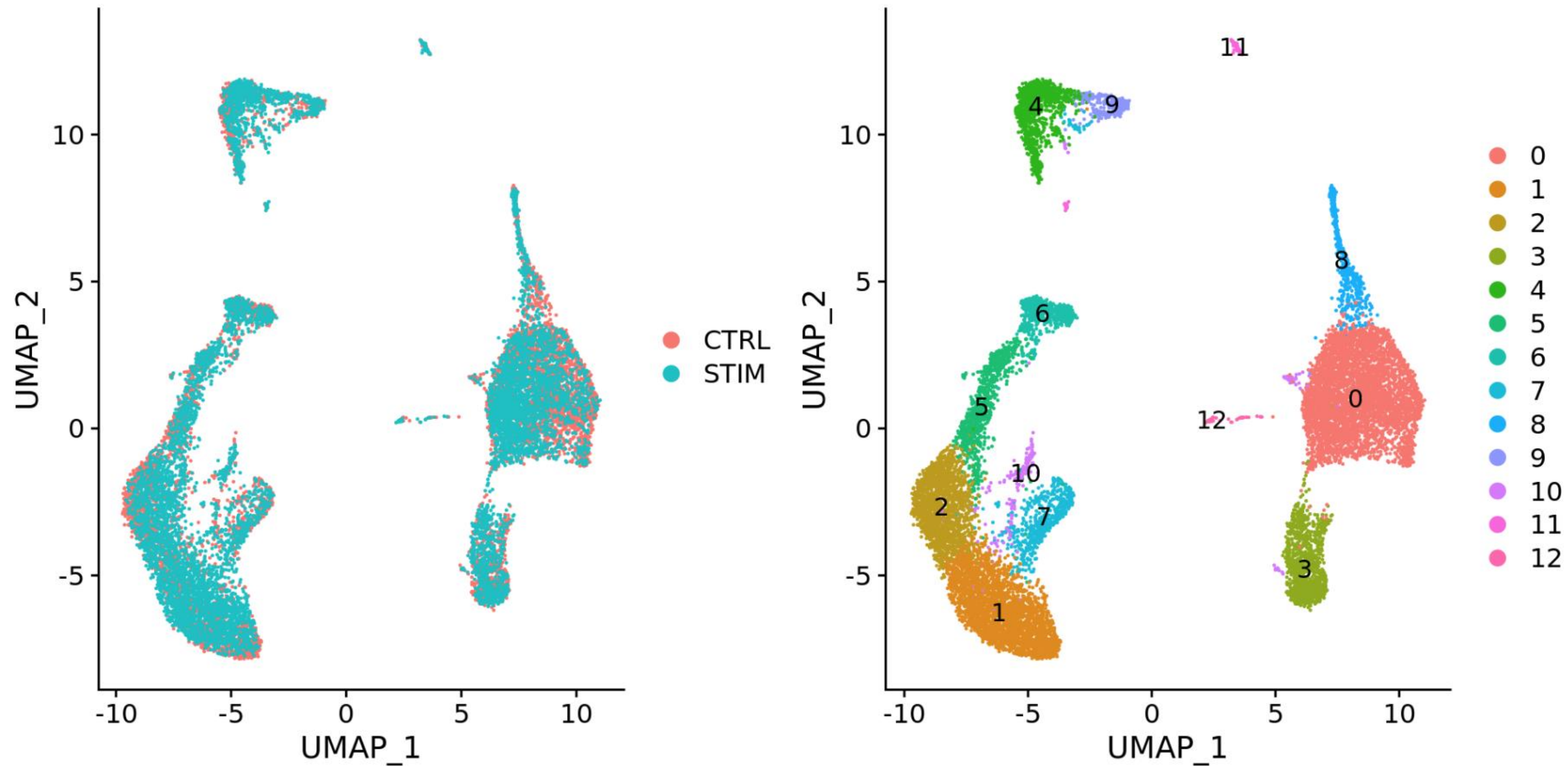


# Exercise: Identify problem in this plot

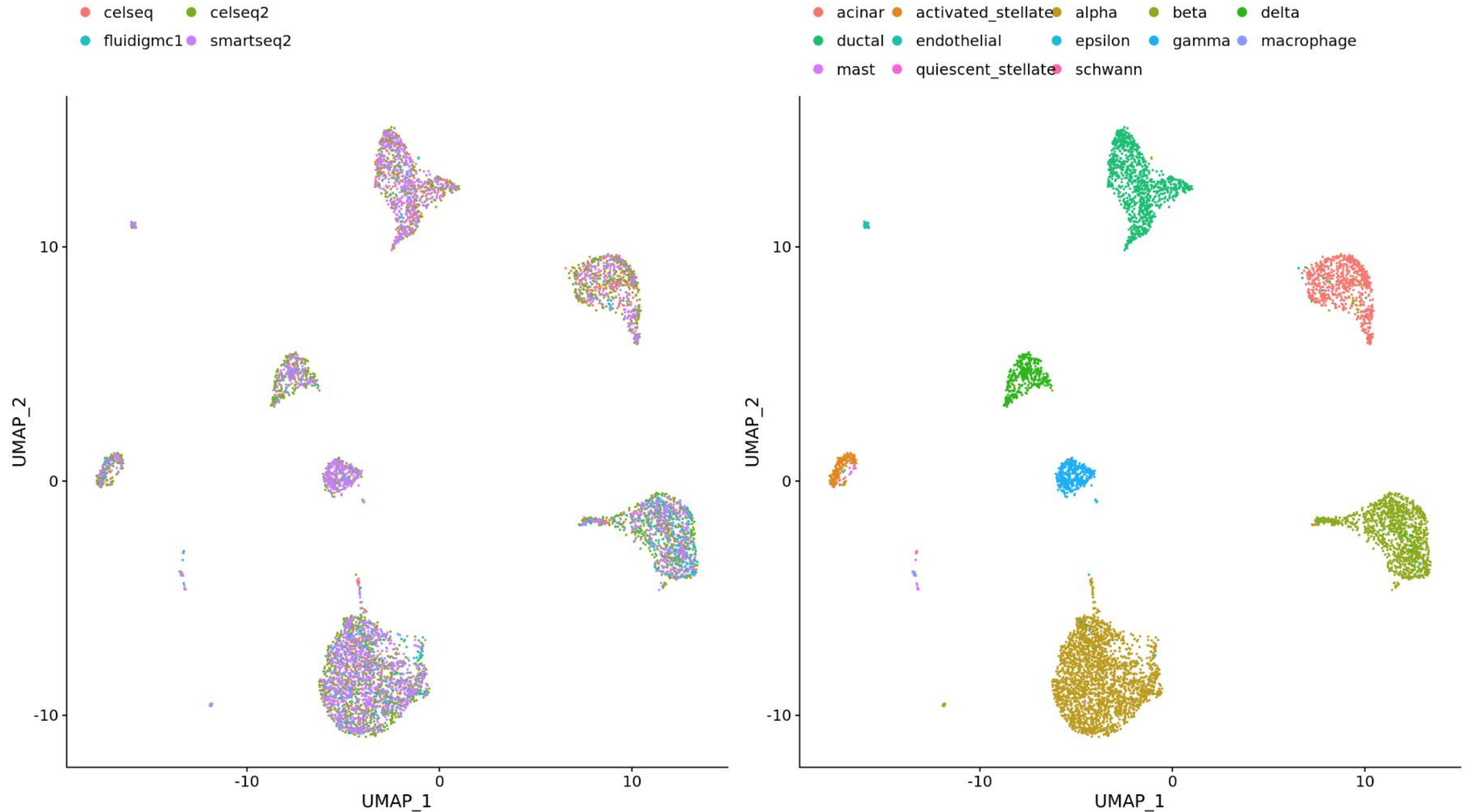


- **Explore the data:** do not just always perform integration because you think there might be differences
- If cells cluster by **sample, condition, batch, dataset, modality**, performing integration can help align cells across the groups to greatly improve the clustering and the downstream analyses.

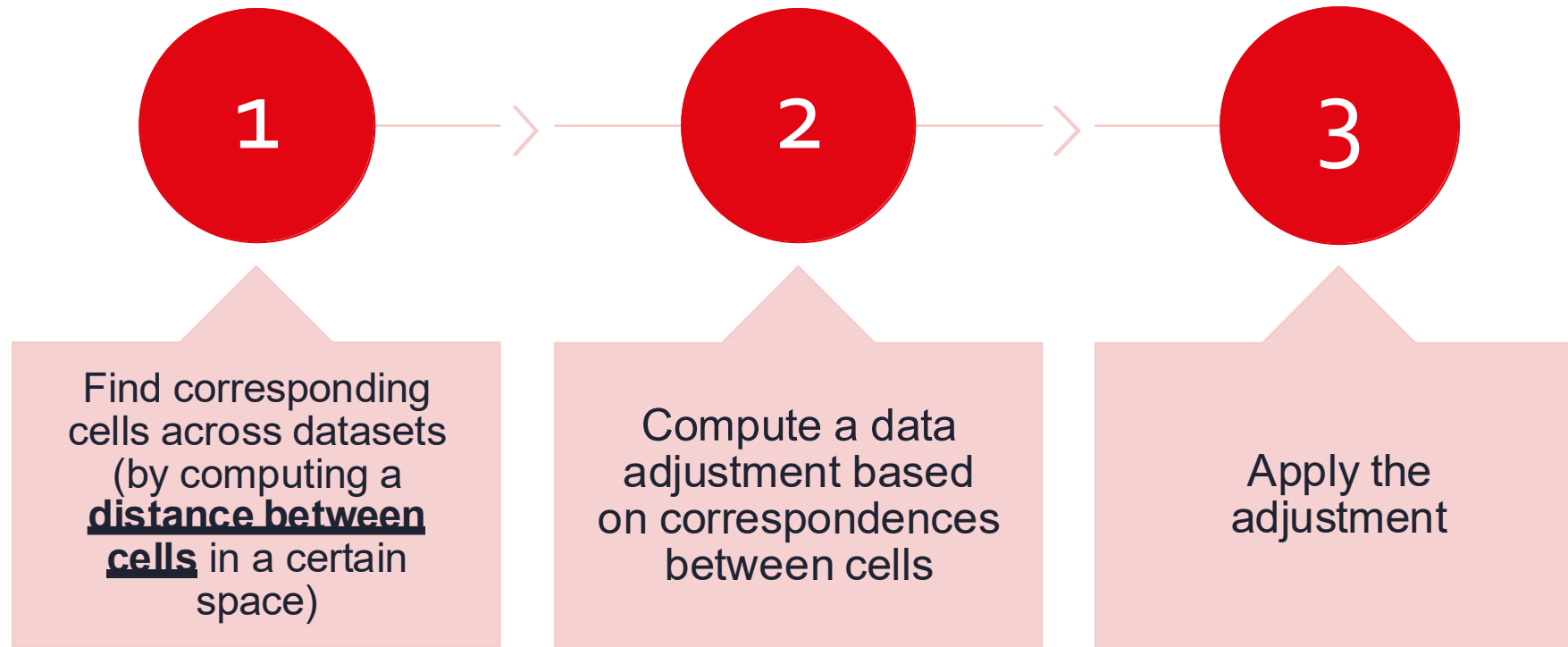
# Example scenarios for integration: conditions



# Example scenarios for integration: datasets

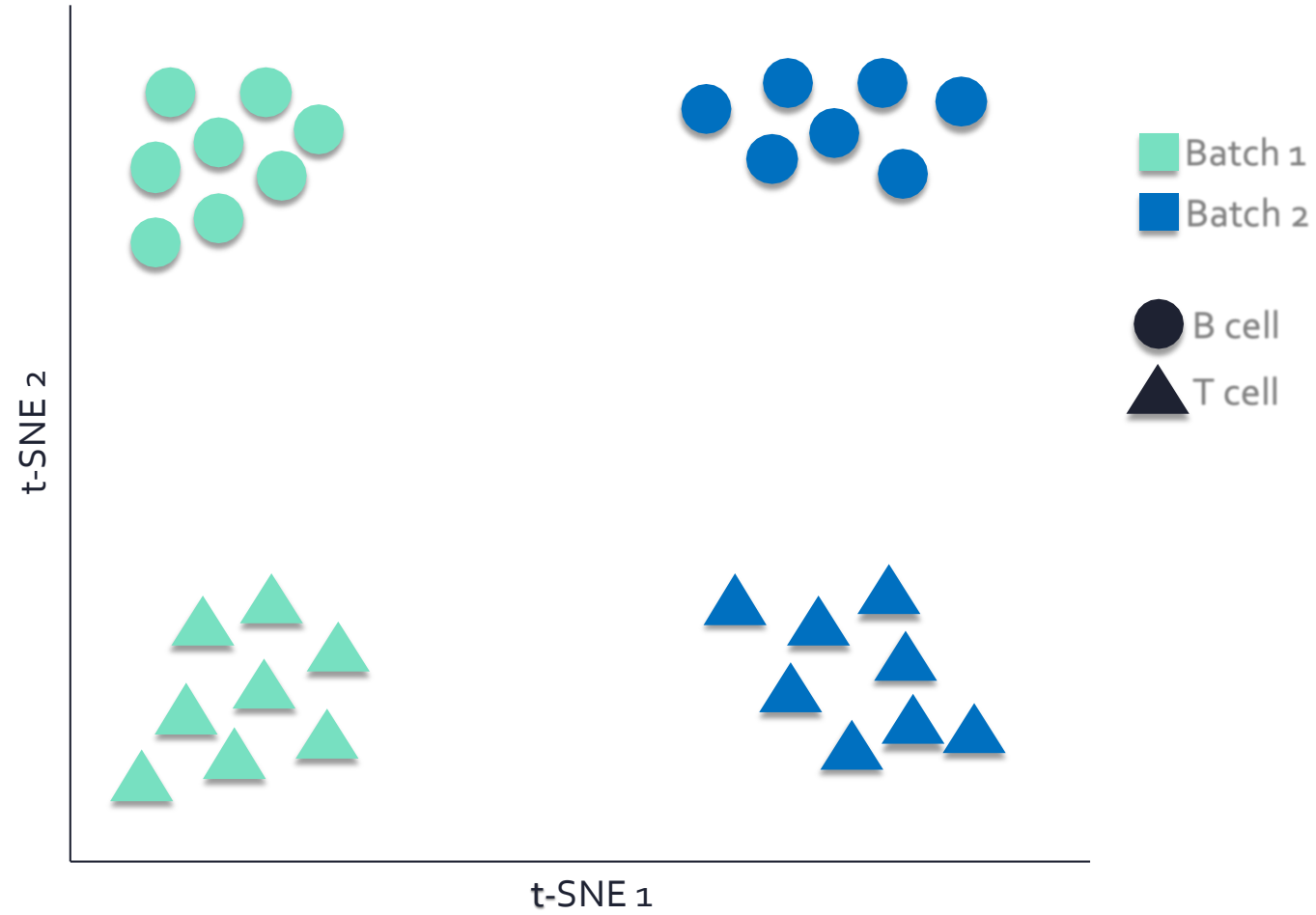


# How to integrate the cells?

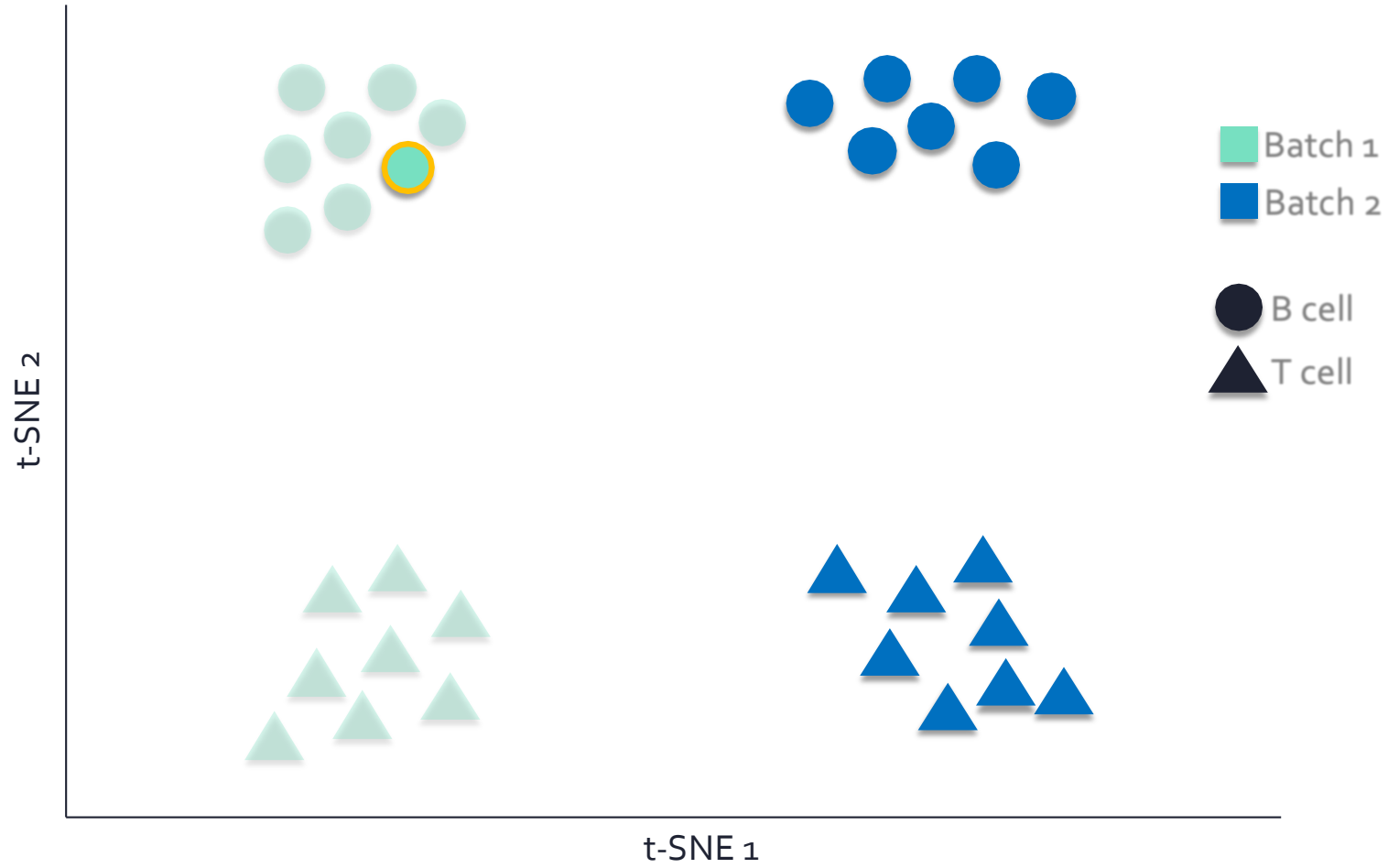


# How to integrate the cells?

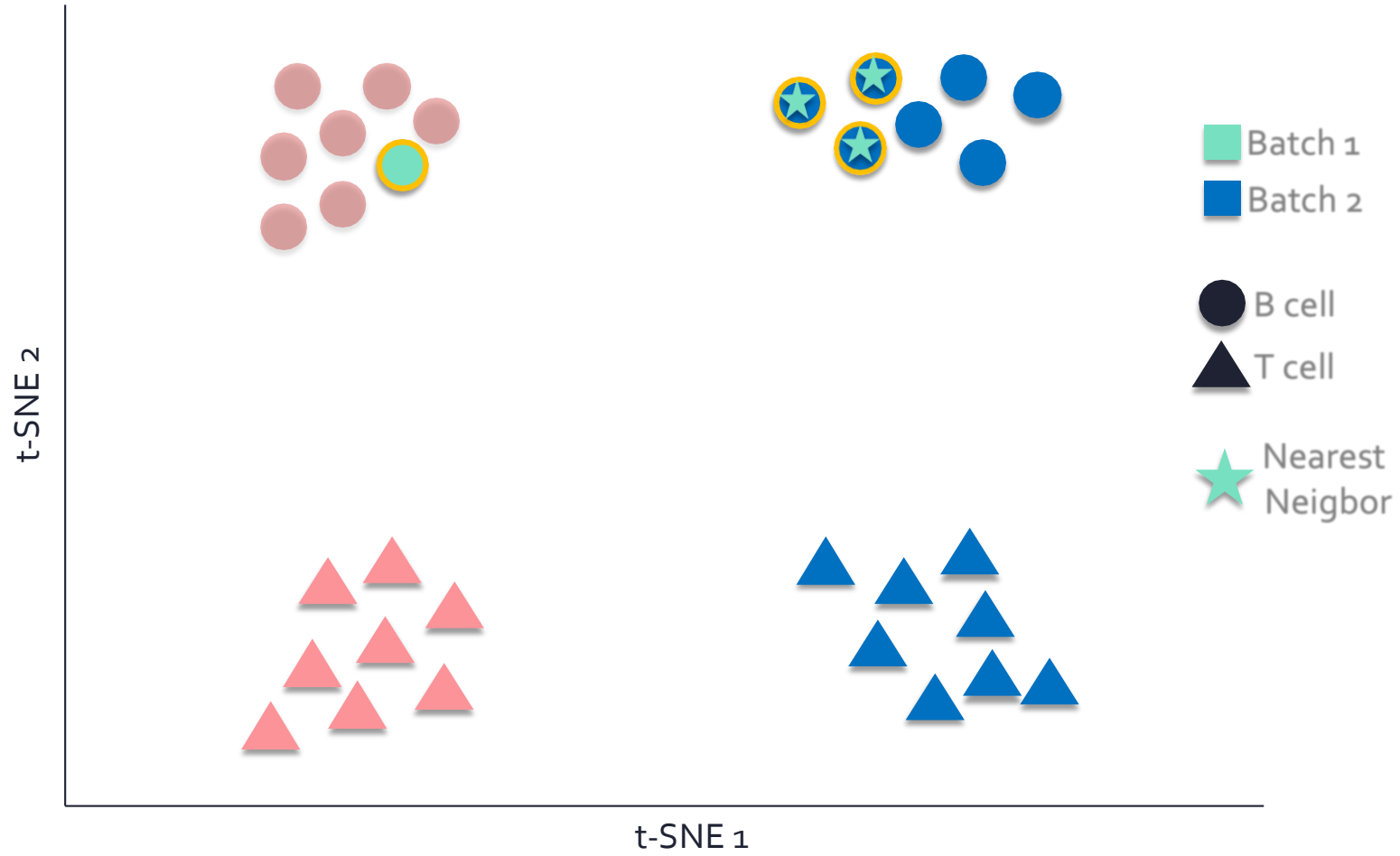
## Example



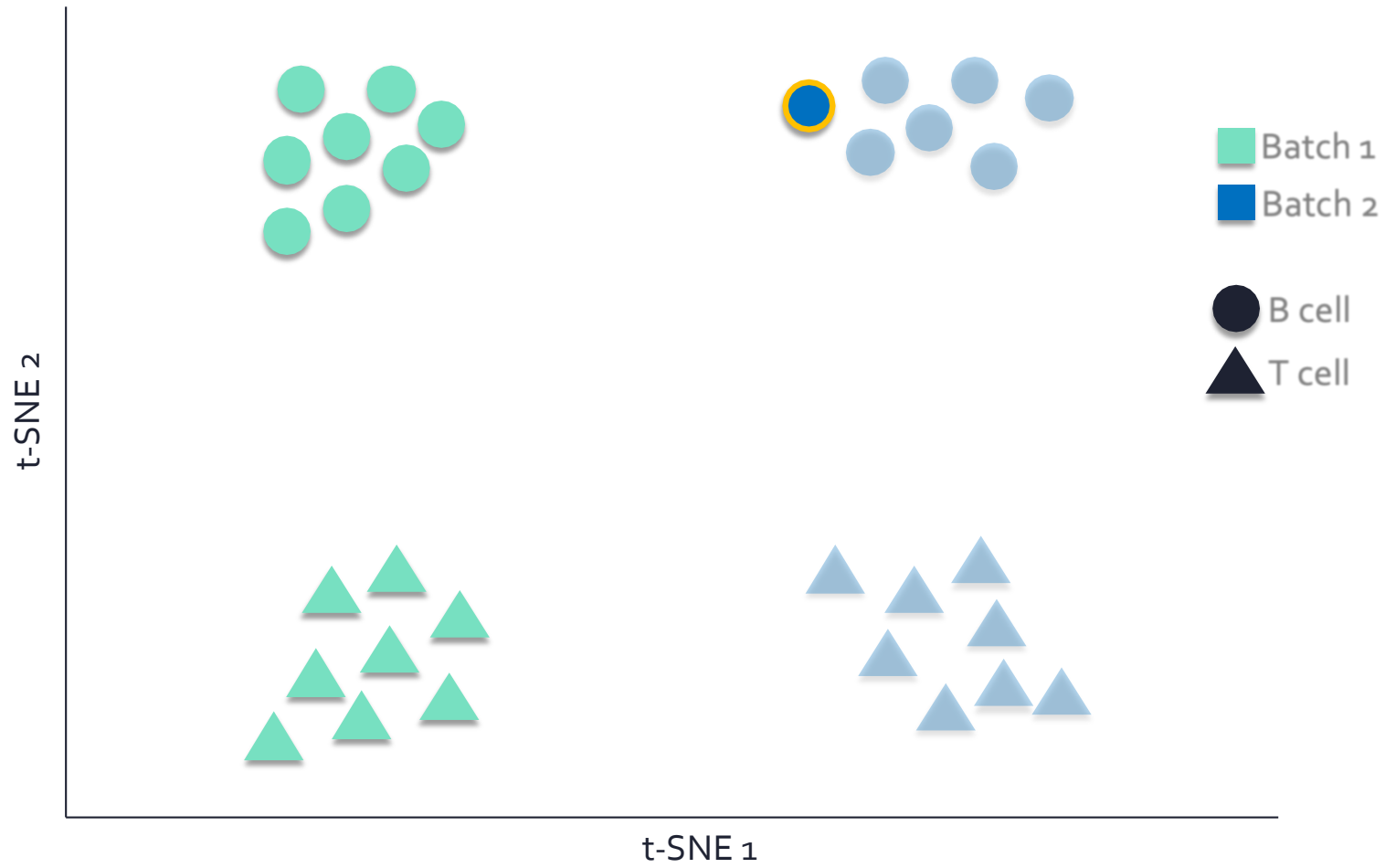
# How to integrate the cells?



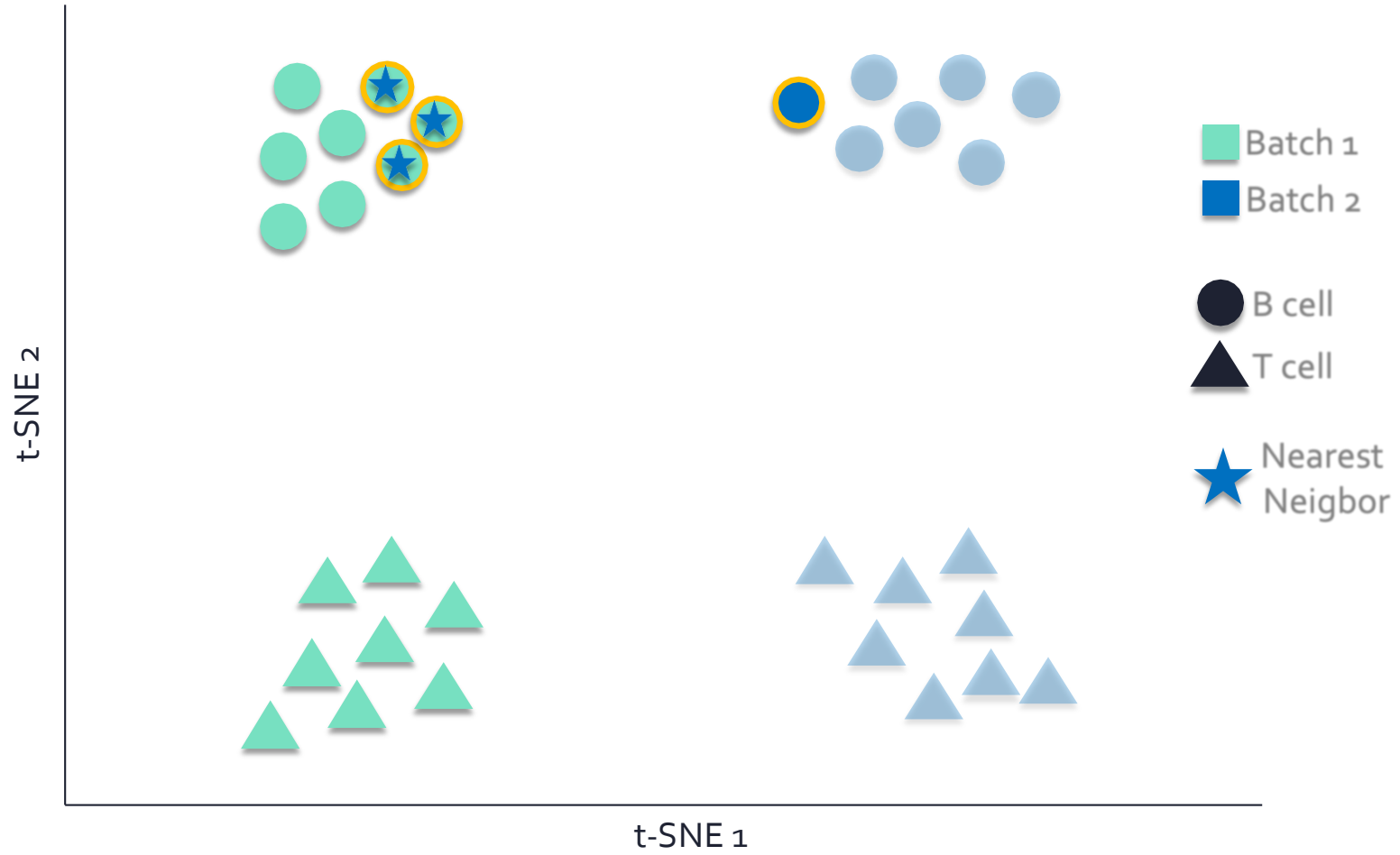
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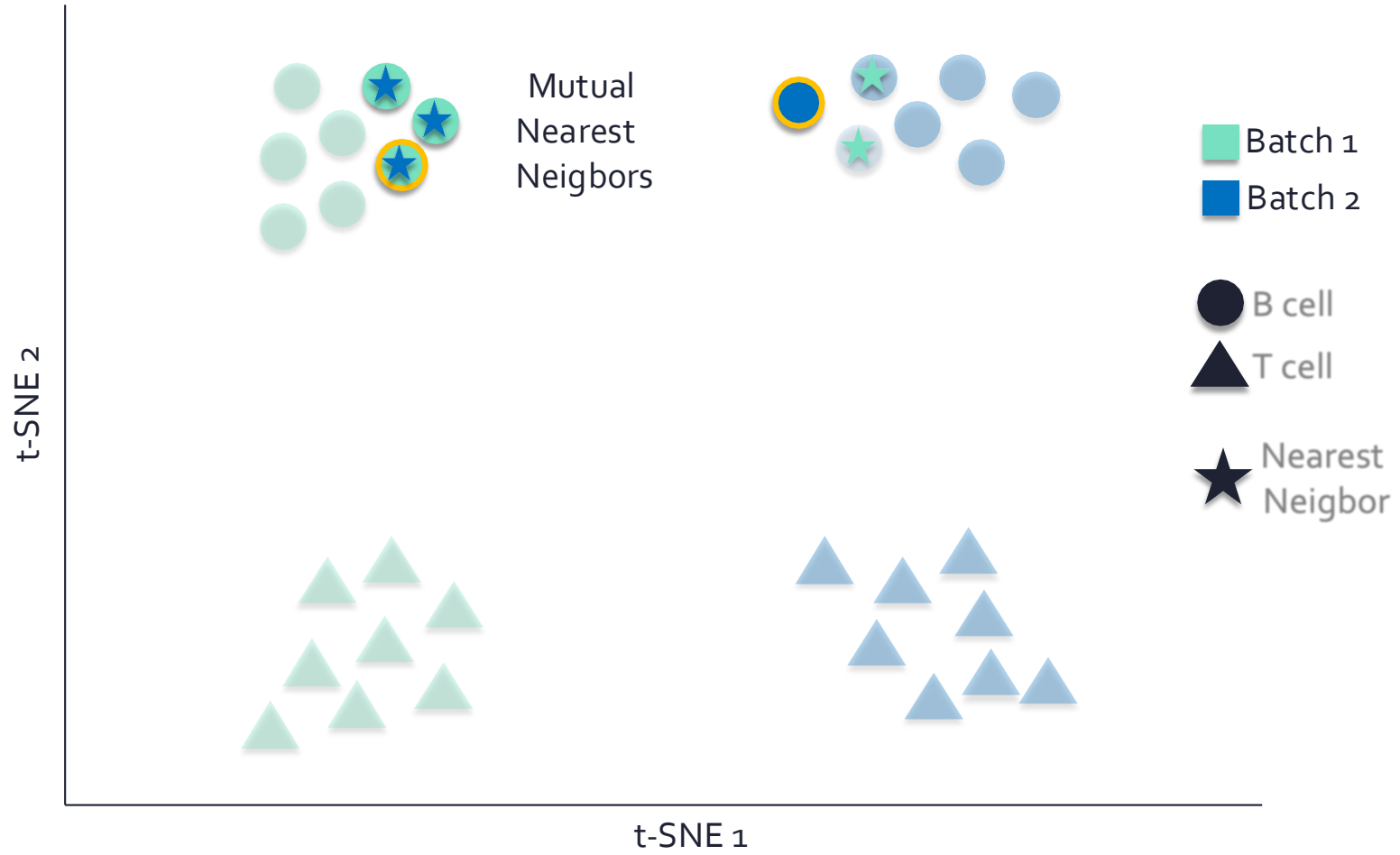
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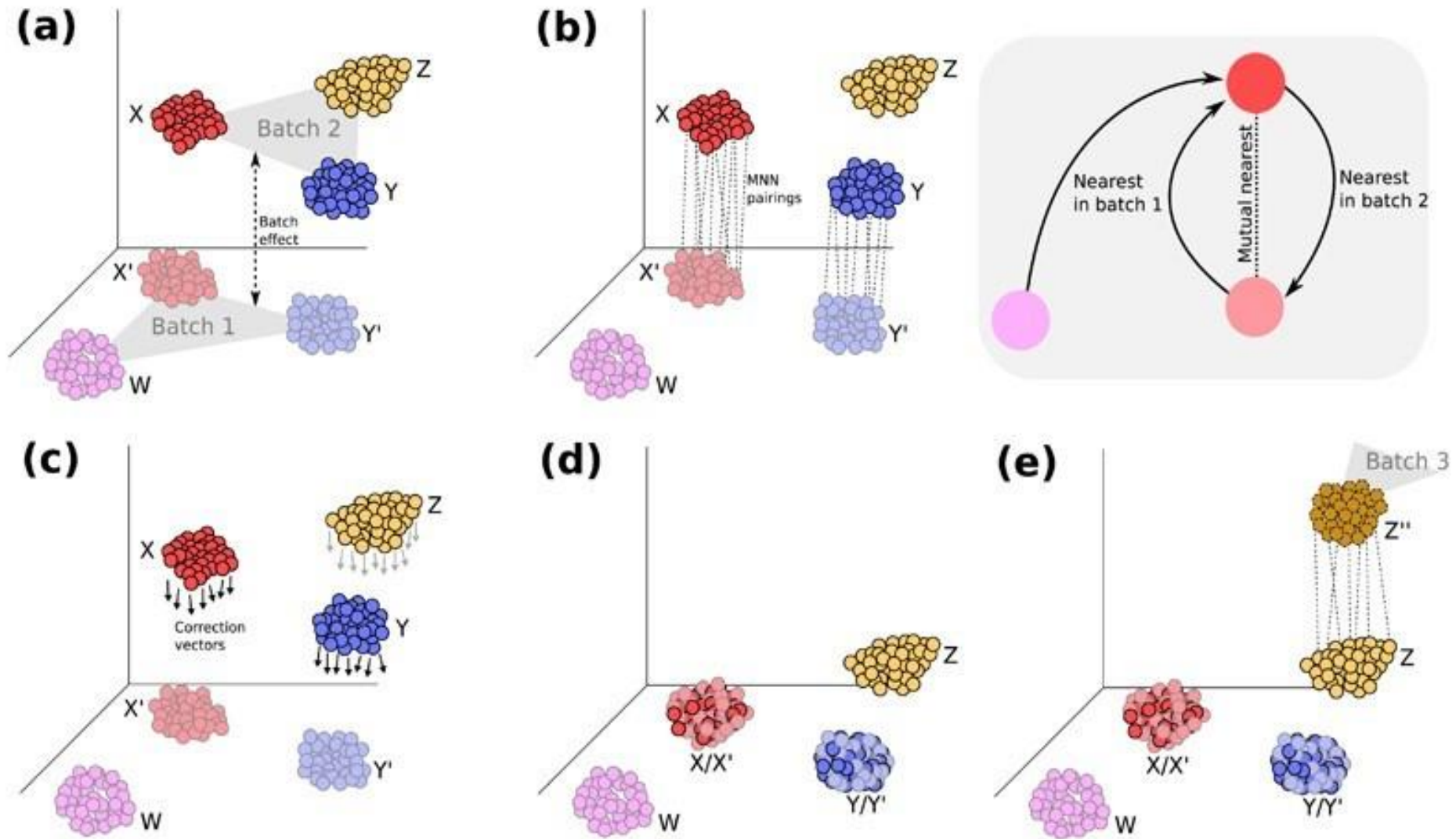
# How to integrate the cells?



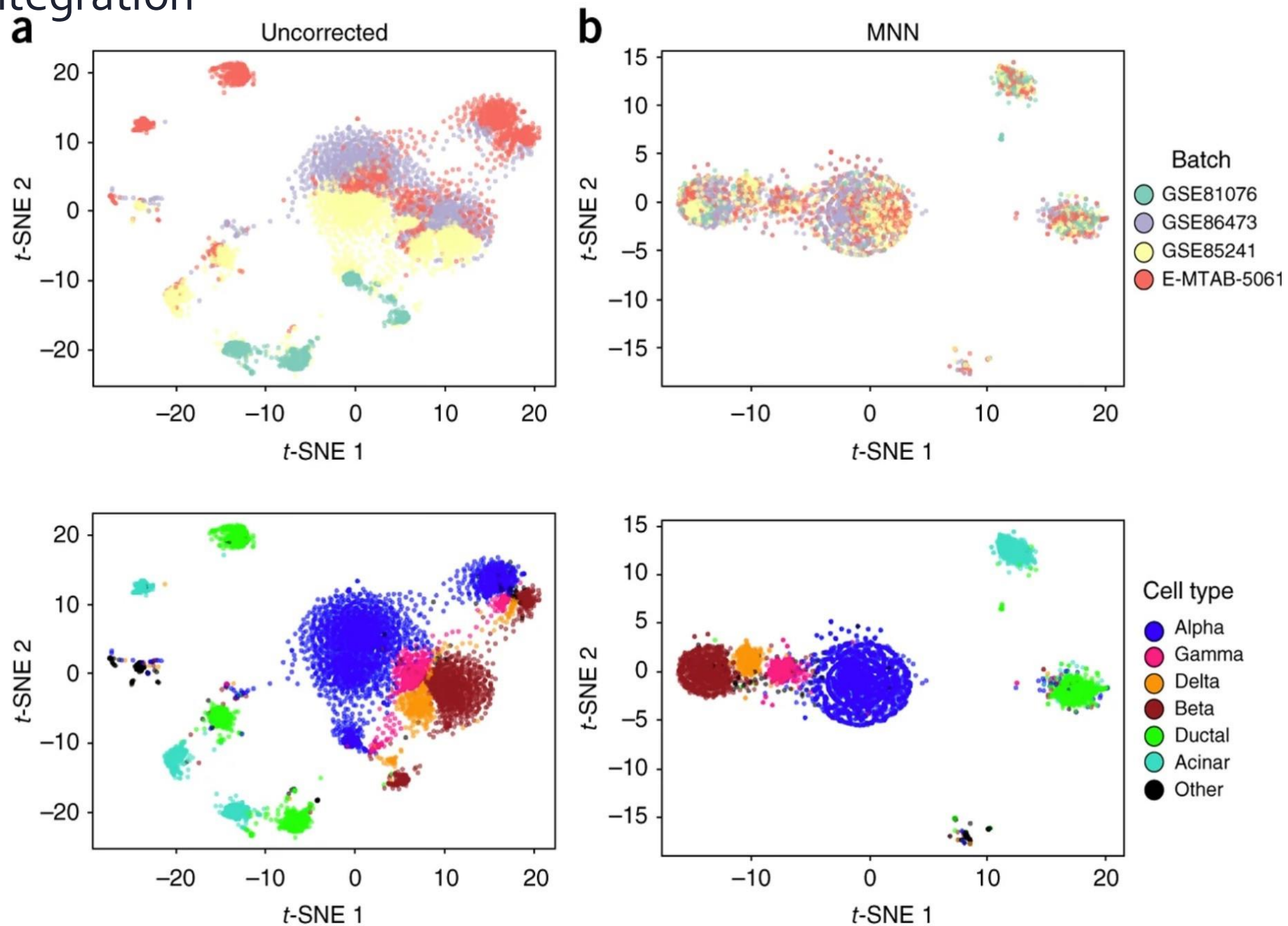
# How to integrate the cells?



# Mutual Nearest Neighbors



# Final Integration



# Canonical Correlation Analysis (CCA) + anchors

## Find

Find corresponding cells across datasets (anchors) in L2-normalized CCA

## Compute

Compute a data adjustment based on correspondences between cells

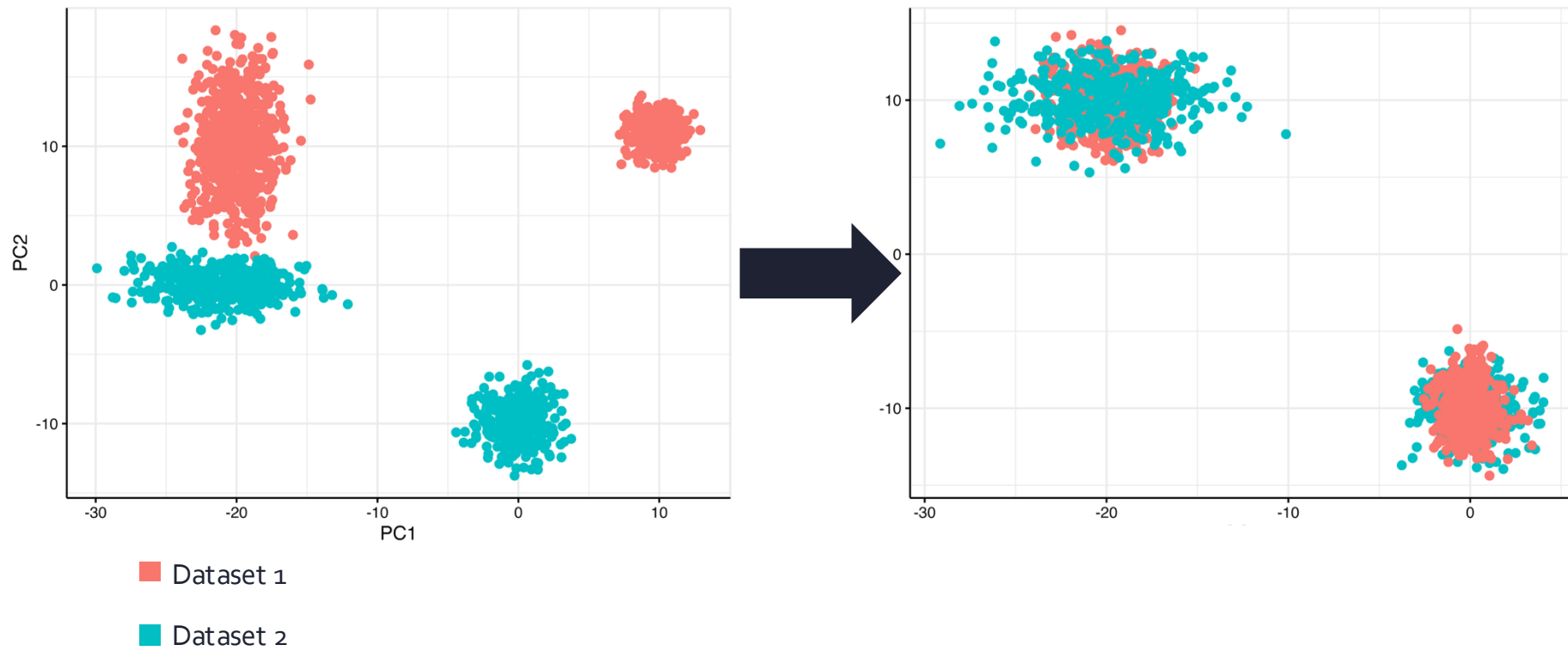
## Apply

Apply the adjustment

# Canonical Correlation Analysis (CCA) + anchors

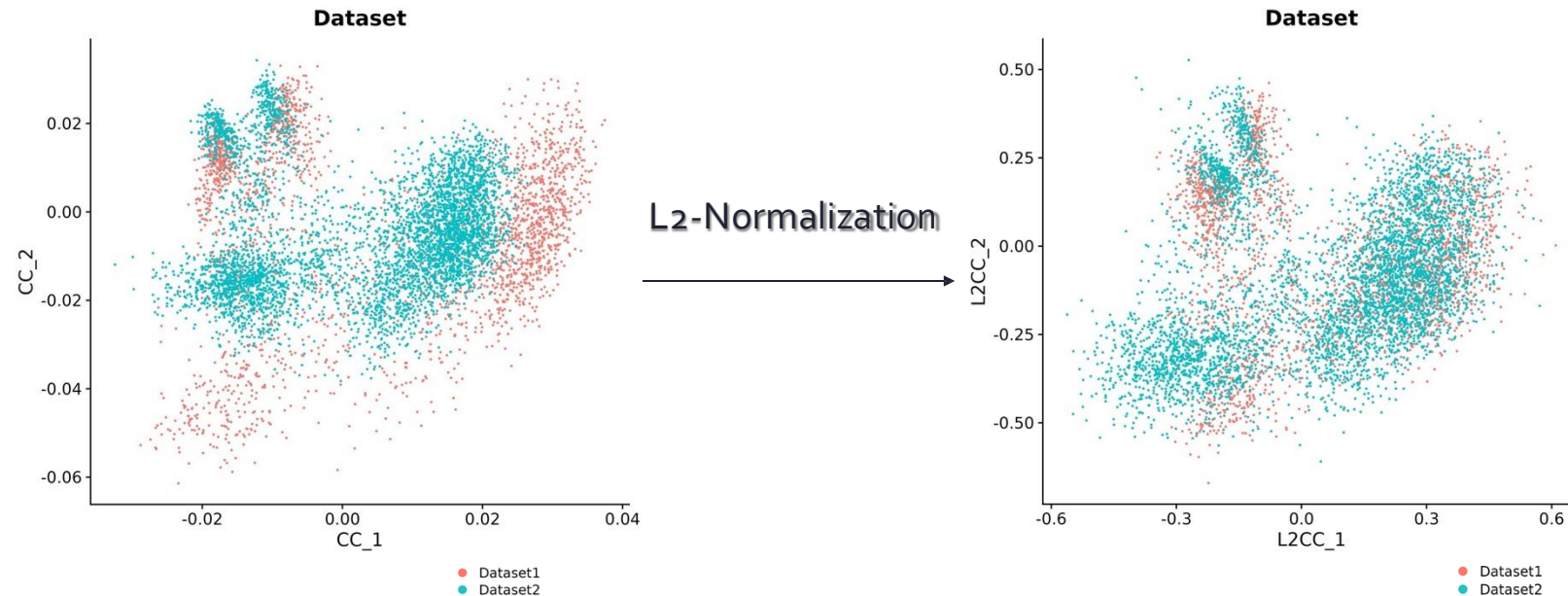
## Step1

Find corresponding cells across datasets



# Canonical Correlation Analysis (CCA) + anchors

## L2-normalization of CCs



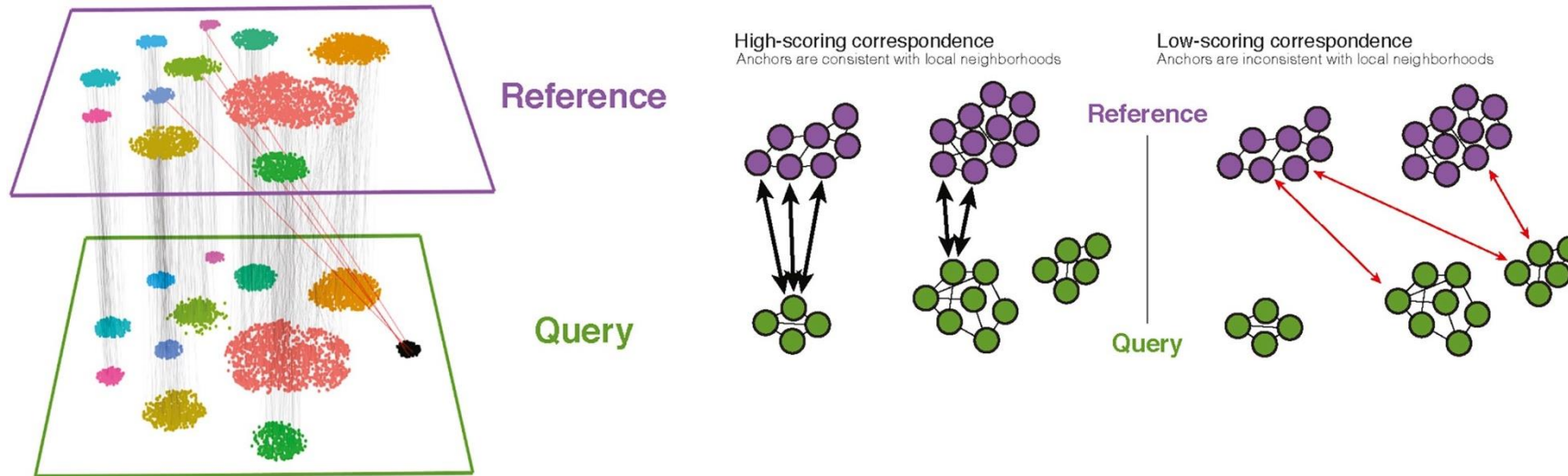
Imagine you have two datasets, A and B, each with a set of genes. CCA tries to find **linear combinations of genes** in A that correlate with corresponding combinations in B.

CCA looks for **pairs of “canonical” variables** (one from each dataset) that are maximally correlated with each other, capturing the shared structure

# Canonical Correlation Analysis (CCA) + anchors

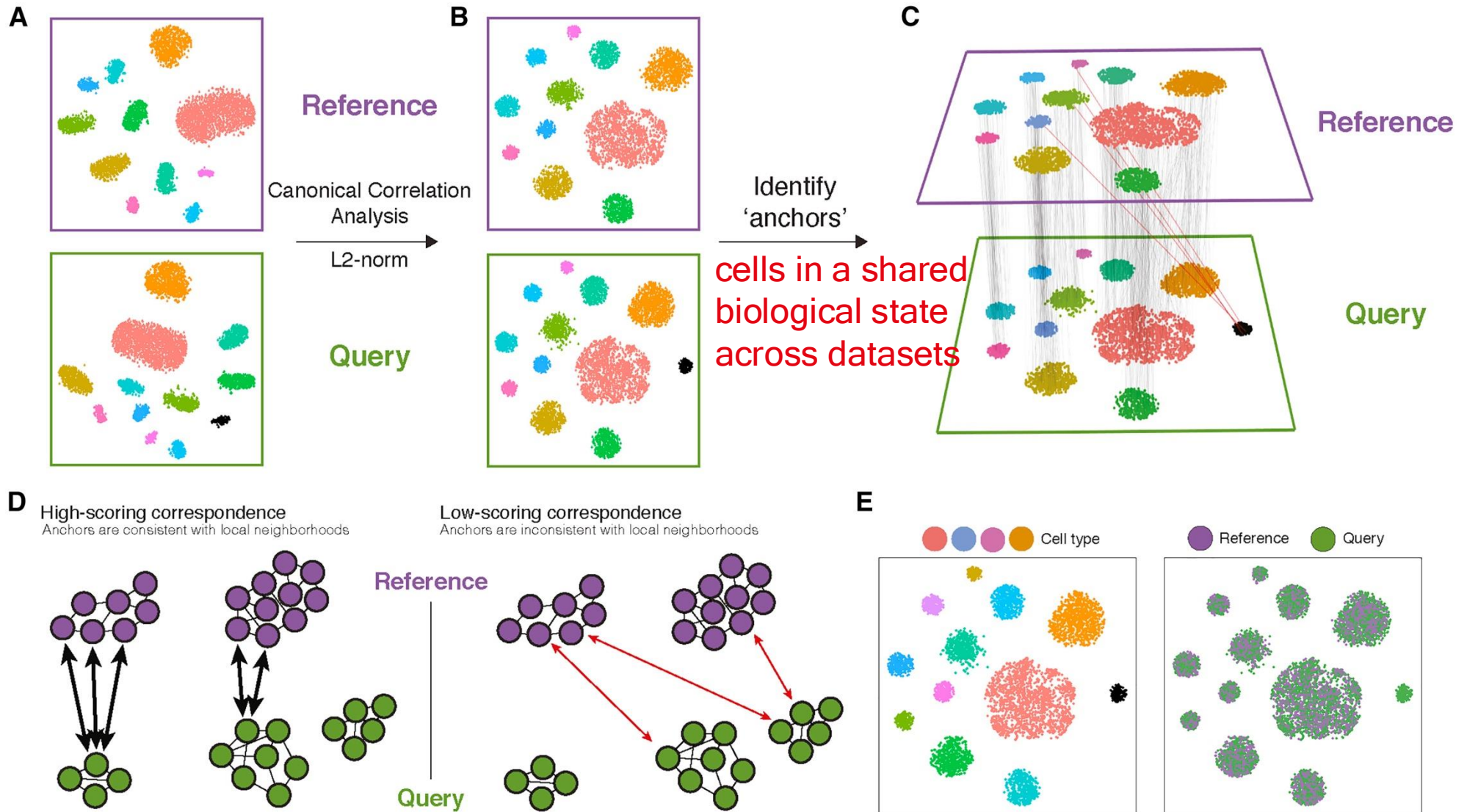
## Anchors Identification

Seurat first performs **mutual nearest neighbours (MNN) matching** to identify cell pairs that are closest in gene expression space across datasets.



# Integration using CCA: canonical correlation analysis

Datasets coming from 2 different platforms



scores to compute "correction" vectors for each query cell,  
transforming its expression so it can be jointly analyzed

# Is CCA in Seurat really a CCA?

The “Seurat CCA” is taking the projection vector from the traditional CCA directly as cell embeddings. But in fact, **the classical definition of CCA would imply projecting genes into a common space rather than cells.**

Based on our understanding, the math behind the “Seurat CCA” algorithm is technically closer to a dual PCA formulation.

In the original paper, the assumption that the covariance matrix of gene expression is diagonal, is not necessary.

Furthermore, considering the formulation to preserve the most similarity (dual PCA), the low-dimensional cell embeddings should multiply the singular value, which is currently missing in the “Seurat CCA” algorithm.

And finally, there is an intrinsic connection between MNN and “Seurat CCA” (extended dual PCA).

[https://xinmingtu.cn/blog/2022/CCA\\_dual\\_PCA/](https://xinmingtu.cn/blog/2022/CCA_dual_PCA/)

# Quiz

**In which condition will you perform integration?**

- A) When cells cluster by sample
- B) When cells cluster by condition
- C) When cells cluster by batch
- D) When cells cluster by dataset
- E) None of the above
- F) All of the above

# Guidelines to choose an integration method

**a**

|              | Considerations                                 | scANVI | Scanorama embed | scVI | FastMNN embed | scGen | Harmony | FastMNN gene | Seurat v3 RPCA | BBKNN | Scanorama gene | ComBat | MNN | Seurat v3 CCA | trVAE | Conos | DESC | LIGER | SAUCIE embed | SAUCIE gene |
|--------------|------------------------------------------------|--------|-----------------|------|---------------|-------|---------|--------------|----------------|-------|----------------|--------|-----|---------------|-------|-------|------|-------|--------------|-------------|
| Input        | Programming language                           |        |                 |      |               |       |         |              |                |       |                |        |     |               |       |       |      |       |              |             |
|              | Method runs without additional information     | ✗      |                 |      |               | ✗     |         |              |                |       |                |        |     |               |       |       |      |       |              |             |
| Scib results | Consistent top performer                       | ✓      | ✓               | ✓    |               | ✓     |         |              |                |       |                |        |     |               |       |       |      |       |              |             |
|              | Top method on small/simple tasks               |        | ✓               |      | ✓             | ✓     | ✓       |              |                |       |                |        |     |               |       |       |      |       |              |             |
|              | Top method on large/complex tasks              | ✓      | ✓               | ✓    |               | ✓     |         |              |                |       |                |        |     |               |       |       |      |       |              |             |
|              | Top method on ATAC data                        | —      |                 | —    |               |       | ✓       |              |                |       |                |        |     |               |       |       |      | ✓     |              |             |
| Task details | Integrates strong batch effects                | ✓      | —               | —    |               | ✓     |         |              | —              | —     |                |        |     | —             |       |       |      |       |              |             |
|              | Top method for recovery cell states or modules | ✓      | ✓               |      |               |       |         |              |                |       | ✓              | ✓      | ✓   |               |       |       |      |       |              |             |
|              | Confounding of bio and batch variance          | ✓      | —               |      |               | ✓     |         |              |                |       |                |        |     |               |       |       |      |       |              |             |
|              | Top method for trajectories                    | —      | ✓               | —    | ✓             | ✓     |         |              |                |       |                |        |     |               |       |       |      |       |              |             |
|              | Method deals with varying compositions         |        |                 |      |               |       |         |              |                |       |                | ✗      |     |               |       |       |      |       |              |             |
| Speed        | Fast method for quick results                  |        |                 |      |               |       |         |              |                | ✓     |                | ✓      |     |               |       |       |      |       |              |             |
|              | Scales well to large datasets on CPU           | ✓      | —               | ✓    |               |       |         |              |                | ✓     | —              |        |     |               |       |       |      |       | ✓            | ✓           |
|              | Method has GPU support                         | ✓      |                 | ✓    |               | ✓     |         |              |                |       |                |        |     |               | ✓     |       | ✓    |       | ✓            | ✓           |
|              | Scales well to feature spaces beyond genes     |        |                 |      |               |       |         |              |                |       |                |        |     |               | ✓     | ✓     |      |       |              |             |
|              | Method shows corrected expression              |        |                 |      |               | ✓     |         | ✓            | ✓              |       | ✓              | ✓      | ✓   | ✓             |       |       |      |       |              | ✓           |
| Output       | Method gives relative cell embeddings          |        |                 |      |               |       |         |              |                | ✗     |                |        |     |               |       | ✗     |      |       |              |             |

✓

 Fulfills the criterion

—

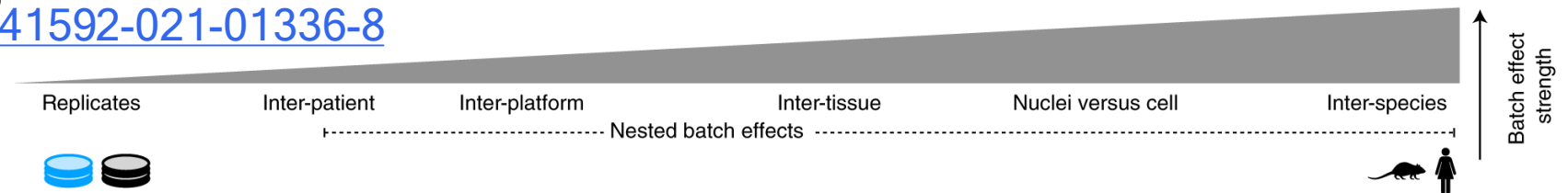
 Partial fulfillment of criterion

✗

 Does not fulfill criterion

Python R

**b**  
<https://www.nature.com/articles/s41592-021-01336-8>



# Summary

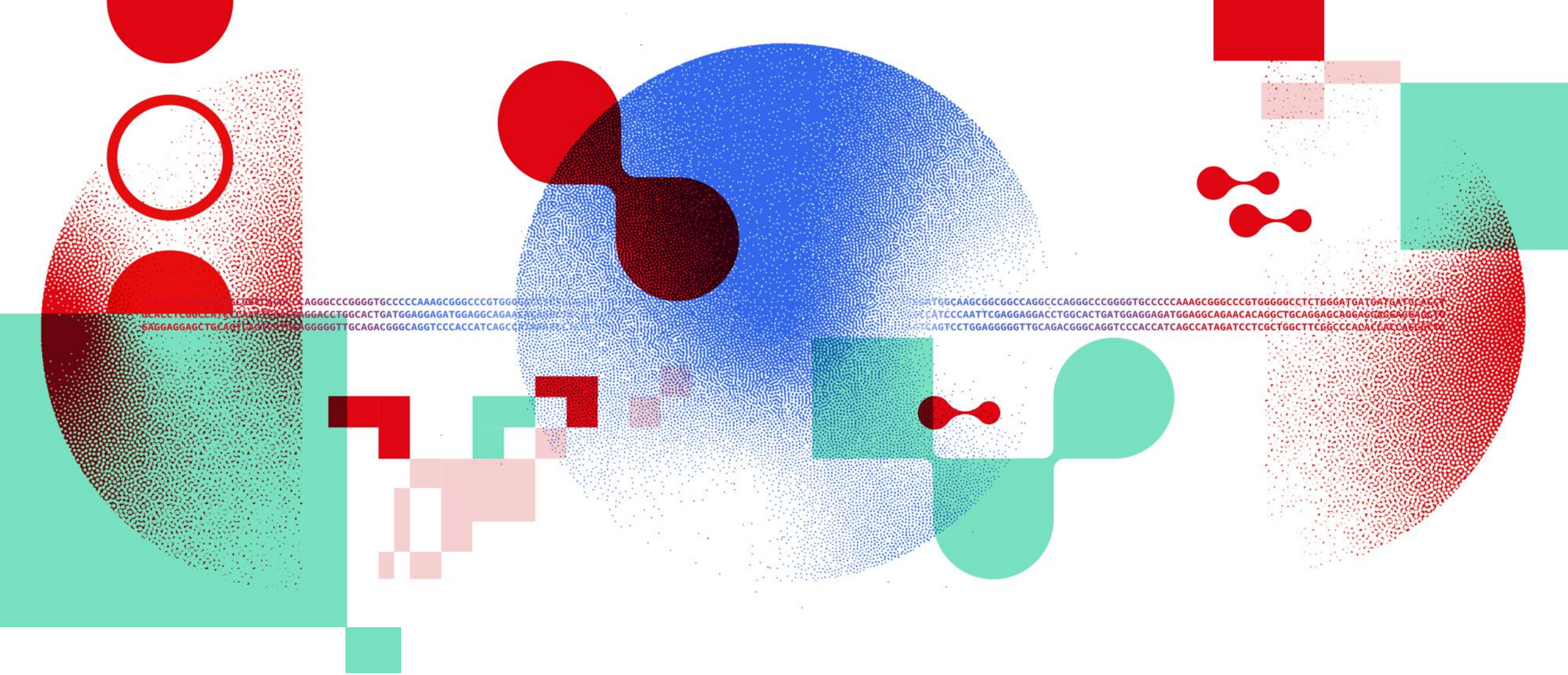
**Experimental Design Matters:** Optimize design to improve data quality and analysis

## **Integration Scenarios:**

- Conditions: Compare different experimental conditions
- Datasets: Analyze data from different platforms together

## **Integration Using CCA:**

- Align cells across groups to improve clustering and downstream analyses
- Compute correction vectors for each query cell to transform its expression for joint analysis



# Thank you

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[sib.swiss](https://sib.swiss)